

FIN Local Research

Checkpoint 000

Baseline State Map Before Programs P428–P430

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Durable recovery point; not a physical-theory claim or final release.

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1 Executive summary

This document freezes the locally auditable FIN research state before any long calculation in the next goal-mode batch. It is a durable recovery point, not a scientific release and not a claim that FIN is a physical theory.

The latest completed local layer is Release 10.35, Programs P411–P427. Its strongest constructive results are exact rational Taylor arithmetic reflected in Lean; a high-precision but ill-conditioned seven-contact KKT candidate; a heralded-erasure-aware symmetric code that beats product and GHZ baselines on the declared finite grid; and a proven variance-optimal positive sampler for twelve Jordan-sampling moments.

The next bounded batch is P428–P430: close or sharply localize the analytic cosine bridge; attempt an interval existence/uniqueness certificate for the P412 KKT candidate; and attempt global dual feasibility by exact or directed-interval Bernstein subdivision.

No selector, physical unit, laboratory record, complete legacy-to-strict bridge, physical-role transfer, total action, Standard Model, gravity theory, or Theory-of-Everything closure is assumed.

2 Epistemic convention

Status	Meaning
[Proven]	Analytic proof, exact arithmetic, proof-kernel result, or exhaustive finite theorem.
[Computer-assisted proof]	Reproducible proof with explicitly verified numerical enclosures.
[Strong evidence]	Stable bounded computation without a full proof certificate.
[Moderate evidence]	Local or model-dependent numerical evidence.
[Conditional]	Requires a named state, reference, calibration, apparatus law, or axiom.
[Refuted]	The declared proposition fails in its stated class.
[Blocked by external evidence]	Independent physical data, calibration, custody, or unblinding is required.

Floating-point residuals are never promoted to theorems. Synthetic records test software only and are never physical evidence.

3 Binding mathematical objects

The historical intermediate kernel and the strict working kernel remain distinct:

$$K_{\text{legacy,ont}}(d) = 4 \ln 2 \frac{\cos(\pi d/4 + \pi/6)}{1 + 0.01d}, \quad K_{\text{strict,gate}}(d) = \frac{\cos((743/4000)d + 13/80)}{1 + d^{9/5}}.$$

The legacy object is an intermediate bridge kernel. The strict object is the later completed/enriched working kernel. No raw identity, source-independent completion map, or physical-role transfer theorem currently identifies their roles.

For a finite self-adjoint generator $A = sI - W$,

$$U_t = e^{-itA}, \quad P_t = e^{-tA}$$

are consequences of one spectral functional calculus. This is mathematical wave/heat duality. It does not supply a physical state, clock, preparation, environment, instrument, detector, apparatus, observer, or record.

4 Current state map

Lane	Current evidence	Boundary	Next
Cosine trust	P411 checks exact degree-40/42 rational arithmetic for twelve phases.	Real cosine remainder theorem remains a named bridge.	P428
Signed-moment optimum	P412 gives a 25-equation, seven-contact candidate with residual 3.42×10^{-28} .	Condition number 2.73×10^{15} ; no interval uniqueness.	P429
Global dual	Contact geometry and old-dual mismatch are known.	New dual lacks a global feasibility certificate.	P430
Transcendence	Injectivity reduces to Lindemann–Weierstrass.	Local formal provider absent.	P431
Damping source	Two named source classes are exactly refuted.	Fractional/subordination/operator-valued classes remain.	P432
Photonic chart	132 directions are full-rank at nine sampled points.	No interval cover or global alias theorem.	P433
Noisy comb	Feasible strategies exist; gap up to 0.52763.	No certified SDP dual.	P435
Erasure code	O149 improves product/GHZ in 16/16 cells.	Numerical, symmetry-restricted, $n \leq 4$.	P436
Twelve modes	No one-use gain over extremal pair on tested grid.	Multi-use/adaptive/nonuniform noise open.	P437
JSR estimator	O151 is exact for the declared loss.	No detector response model.	P440
Orientation	O153 is nonzero and inversion odd.	Sign uses a supplied chart orientation.	P442
Physics	Executable specifications exist.	No laboratory record or dimensional standard.	external

5 Closed and repetition-gated lanes

The following routes remain closed without a genuinely new typed source, theorem, provider, invariant, or blocker-cut:

- generic legacy-to-strict bridge replay;
- silent transfer of legacy physical roles;
- $\text{Aut}(\mathbb{Z}_{12})$ -invariant scalar selector searches;
- radial inversion-odd receiver inventories;
- entropy, damping, amplitude, or scale used as a selector;
- dimensionless ratios promoted to physical units;
- proxy Lagrangian residuals promoted to a full tensor action;
- synthetic data promoted to laboratory evidence;
- repeated finite scans after a theorem-grade no-go.

The ontology remains

$$\text{nadsoliton} \rightarrow \text{light} \rightarrow \text{matter} \rightarrow \text{emergent observer},$$

with no informational layer beneath the nadsoliton.

6 Selected batch P428–P430

6.1 P428: analytic cosine bridge

Question. Can the exact rational intervals of P411 be connected to the real cosine inside the locally available formal stack?

Success. A proof-assistant theorem, with no FIN-specific analytic axiom, showing that the P411 sums enclose real cosine at all twelve phases.

Boundary. If the stack lacks real analysis, export a machine-checkable dependency certificate and a separate exact analytic proof; do not call an assumed bridge internally proved.

6.2 P429: interval KKT certificate

Question. Does a unique exact zero of the 25-equation KKT map exist near O144?

Success. Interval Krawczyk, interval Newton, or radii-polynomial inclusion on a declared box, together with weight-sign and node-order checks.

Boundary. A proved exclusion, singular enclosure, or failure of interval contraction is valid. A smaller floating-point residual is not.

6.3 P430: global dual feasibility

Question. Does the P429 dual polynomial obey $-1 \leq p(x) \leq 0$ on $[0,1]$ with the declared contacts?

Success. Exact rational or outward-rounded Bernstein subdivision outside certified contact boxes, with interval Taylor bounds inside them.

Boundary. One certified violation refutes the candidate. Sampling or plotting alone cannot prove global feasibility.

7 New-object policy

No object name will be introduced unless it closes a typed interface. Possible outputs are limited to a cosine analytic-provider certificate, an interval KKT isolating box, a global contact-aware Bernstein certificate, or a rigorous obstruction explaining why one cannot be completed locally.

Each object must state its domain, codomain, premises, transformation law, necessity, removal test, kernel relation, selector dependence, dimensional status, operational meaning, and failure mode.

8 Explicit nonclaims

This checkpoint does not claim FIN is a physical theory, QW-2191 discharge, a non-premise selector, a canonical phase origin, a physical unit, a completed legacy-to-strict bridge, legacy physical-role transfer, laboratory realization, a Born rule, Lorentz symmetry, Standard Model, gravity theory, L_{total} , or Theory of Everything.

9 Falsification plan

For P428, inspect parity and domain assumptions of the alternating Taylor theorem and attempt boundary counterexamples. For P429, recompute at increased precision, scale variables, estimate interval-condition growth, test several box radii, and require inclusion. For P430, separate contact from noncontact regions, compare power and Bernstein bases, and search for certified sign violations before attempting a global pass.

10 Artifact inventory

The authoritative prior layer is:

- `FIN_Programs_411_427_Reference_Erasure_Report_EN.pdf`;
- `FIN_Programs_411_427_Results.json`;
- `FIN_Programs_411_427_Taylor_Cosine.lean`;
- `fin_programs_411_427.py`;
- `test_fin_programs_411_427.py`;
- `FIN_PROGRAMS_411_427_RELEASE_MANIFEST.sha256`;
- `RELEASE_10_35_PROGRAMS_411_427.md`.

11 Restart instructions

From the repository root:

```
python3 -m unittest -v test_fin_programs_411_427.py
.elan/toolchains/leanprover--lean4---v4.28.0/bin/lean \
  FIN_Programs_411_427_Taylor_Cosine.lean
lualatex -interaction=nonstopmode -halt-on-error \
  FIN_Local_Research_Checkpoint_000_EN.tex
python3 -m unittest -v test_fin_goal_checkpoint_000.py
```

Then begin P428 by auditing which real-analysis definitions and theorems are actually available locally. Do not download a library.

12 Ranked continuation

Rank	Program	Decisive output	Prior
1	P429	Interval existence/uniqueness or rigorous failure	0.72
2	P428	Internal cosine theorem or dependency boundary	0.68
3	P430	Global Bernstein certificate or violation	0.61
4	P436	Certified positive O149 gain in one cell	0.66
5	P440	Apparatus-aware minimax JSR law	0.64
6	P435	Noisy-comb SDP primal/dual certificate	0.58

The immediate order is P428 $\rightarrow$ P429 $\rightarrow$ P430. No expensive next batch begins until a separate P428–P430 PDF is compiled and verified.

13 Baseline conclusion

The most defensible current interpretation is a finite spectral-information and operational mathematical framework with exact theorems, computer-assisted certificates, and explicit experimental interfaces. The missing passage to physics is a typed extension by state, preparation, clock, reference frame, environment, instrument, apparatus response, dimensional standard, and independent record. The next batch attacks mathematical trust boundaries, not physical closure.